

Simultaneous Blind Estimation of Antenna Mutual Coupling and Direction of Arrival

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Abstract—In this paper, we propose a blind estimation method for the antenna mutual coupling matrix and the direction of arrival. First of all, the higher order cumulants based blind estimation method is used to calculate the steering vector of the array. The estimated steering vector can differ from the true steering vector by a factor of -1 or by exchanging of the real and imaginary parts. These differences as well as the direction of arrival and the mutual coupling matrix are estimated afterwards using the least square error criteria. Simulations show the proposed algorithm can converge to the exact solution very quickly and is suitable for the closely aligned antenna arrays for the massive multiple input and multiple output (MIMO) systems.

Keywords—Antenna; Mutual coupling; MIMO

I. INTRODUCTION

With the increasing demand from the wireless communication networks, the massive multiple input and multiple output (MIMO) technique has gained significant attentions among the industrial and research communities [1, 2]. The massive MIMO technology employs a large number of closely arranged antennas for the transmitters and the receivers. The diversity gain and the multiplexing gain from the antenna array will greatly increase the overall system capacity. Due to the presence of the mutual coupling between the closely aligned antenna elements [3], the existing channel estimation algorithm might not function well while ignoring the coupling effect. For example, while employing the multiple signal classification (MUSIC) algorithm [4] or estimation of signal parameter via rotational invariance technique (ESPRIT) algorithm [5] to estimate the direction of arrival (DOA) of the signals, the coupling matrix will modify the signal space and therefore degrades the accuracy of the algorithm. This becomes even more severe in millimeter wave massive MIMO systems, which employs highly integrated millimeter antenna arrays with stronger mutual coupling effect. An algorithm that can recover the correct DOA of the signal as well as the coupling matrix of the antenna array will be greatly useful for the closely arranged antenna arrays.

There have been great efforts to characterize the mutual coupling effect during the estimation of DOA. Marc Mowla et al. [6] proposed a measurement approach to address the

coupling among the arrays as well as the phase measurement errors among the antenna elements. The approach, however, require a pre-defined signal incoming direction and all the estimation are dependent on the accuracy of the known direction. This might be helpful during the base station set-up stage, but is impractical during the operation of the wireless system. Furthermore, it is revealed that the mutual coupling between the antenna elements may vary when the signal comes from the different directions. The method introduced in [6] becomes inapplicable to track the variation of the mutual coupling matrix in such situations. T. T. Zhang et. al. [7] proposed an algorithm to estimate the mutual coupling matrix and the DOA based on the known steering vector of the array. However, the measurement of the steering vector is also difficult in the operational wireless systems. To summarize, despite the pioneering works in [6] and [7], there is a great need for an algorithm that can achieve simultaneous blind estimation of the mutual coupling matrix and the signal DOA.

In this paper, we propose a novel algorithm to blindly estimate mutual coupling and signal DOA without any prior information. The procedure of the algorithm is as follows: the steering vector of the antenna (with the impact of the mutual coupling) is estimated by the higher order cumulants of the received signals [8]. The estimated steering vector is consistent with the real steering vector despite a factor of -1 or an exchange of the real and imaginary parts. These slight differences as well as the mutual coupling matrix and the signal DOA are estimated using the least square error (LSE) method. The main contributions of this work are to combine the algorithms in [7] and [8] and to develop an algorithm that can estimate the mutual coupling and the DOA simultaneously. Numerical simulations are provided to verify the proposed algorithm.

II. ALGORITHM PROCEDURES

A. Signal model

The input signal and the received signal are related by

$$\begin{aligned}\mathbf{x} &= \mathbf{B}\mathbf{s} + \mathbf{n} \\ \mathbf{B} &= \mathbf{C}\mathbf{A}\end{aligned}\tag{1}$$

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where $\mathbf{A}$ stands for the steering vector composed matrix without considering the antenna mutual coupling effect,

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ e^{-j\frac{kd}{2}\sin\theta_1} & e^{-j\frac{kd}{2}\sin\theta_2} & & e^{-j\frac{kd}{2}\sin\theta_N} \\ & & & \\ e^{-j(M-1)\frac{kd}{2}\sin\theta_1} & e^{-j(M-1)\frac{kd}{2}\sin\theta_2} & & e^{-j(M-1)\frac{kd}{2}\sin\theta_N} \end{bmatrix}$$

$\mathbf{C}$ the antenna mutual coupling matrix, $\mathbf{B}$ the real steering vector composed matrix, which is to be estimated using algorithm based on the higher order cumulants, $\mathbf{s}$ the signal vector, whose elements are independent variables, $\mathbf{n}$ the noise vectors, whose elements are random Gaussian variables, $\mathbf{x}$ the received signals, M the number of antennas, N the number of signal sources, k the wave number, d the spacing between the adjacent antenna elements.

B. Estimation of the steering vector using higher order cumulants of the received signals

Most of the algorithms to estimate the DOA and the antenna mutual coupling matrix rely on the accurate information of the steering vector [7]. This information can be obtained via measurement as indicated in [7]. However, such measurement is only applicable in the set-up stage while calibrating the antenna array. As discussed in the introduction section, the mutual coupling, however, varies during the operation of the antenna array. Therefore, real time monitoring of the steering vector is in great need.

Blind signal processing algorithms have been widely used to estimate the MIMO channel parameters without any prior knowledge about the input signal and the channel itself. The estimation algorithms are usually based on the assumption of the independent input signal sources. This assumption is also a prior requirement of the DOA estimation algorithms such as MUSIC and ESPRIT, and is often fulfilled. In the case of correlated input signal sources, the assumption can be fulfilled after spatial smoothing of the received signals [9].

There are several widely used blind signal estimation algorithms, including the independent component analysis (ICA) [10] and the higher order cumulants based algorithm [8]. ICA aims to estimate the inverse matrix of the MIMO channel [10] and therefore, it is not straight forward for the steering vector estimation.

The higher order cumulants based algorithm is based on the real signal processing technique [8] while the steering vector is a complex vector. Hence, the algorithm needs to be carefully designed before implementing for the estimation of the steering vectors. In this section, the algorithm is first described by assuming that the steering vectors are real [8]. Afterwards, the algorithm is modified to incorporate the complex nature of the steering vector.

The algorithm for a real MIMO channel matrix estimation can be found in [8], for readers' convenience, it is described as follows:

Step 1, calculate the correlation matrix of the received signal $\mathbf{x}$. Without loss generality, the input signal $\mathbf{s}$ is assumed to have unit power. Hence, the correlation matrix of the received signal vector $\mathbf{x}$ can be calculated as [8]

$$\begin{aligned} \mathbf{R}_{xx} &= \mathbf{B}\mathbf{R}_{ss}\mathbf{B}^T + \mathbf{R}_{nn} \\ &= \mathbf{B}\mathbf{B}^T + \sigma^2\mathbf{I} \end{aligned} \quad (2)$$

$\mathbf{R}_{xx}$ is a Hermitian matrix, while $\mathbf{B}$ is real, which is real symmetric matrix and can be decomposed into [8]

$$\begin{aligned} \mathbf{R}_{xx} &= \mathbf{U}\mathbf{D}\mathbf{U}^T \\ \mathbf{U} &= [\mathbf{U}_s \ \mathbf{U}_n] \\ \mathbf{D} &= \begin{bmatrix} \mathbf{D}_s & \\ & \mathbf{D}_n \end{bmatrix} \end{aligned} \quad (3)$$

where $\mathbf{U}$ is an orthogonal matrix, $\mathbf{D}$ a diagonal matrix. $\mathbf{U}$ is separated into two parts. $\mathbf{U}_s$ is formed by the eigen vectors related to signals and $\mathbf{U}_n$ is formed by the eigen vectors related to the noise. The largest N elements of $\mathbf{D}$ is related to the signals, and their corresponding eigen vectors are the rows of $\mathbf{U}_s$, while the smaller $M-N$ elements of $\mathbf{D}$ is related to noise. Hence, $\mathbf{B}$ should be related to $\mathbf{U}_s$ by [8]

$$\begin{aligned} \mathbf{B} &= \mathbf{U}_s\sqrt{\mathbf{D}_s}\mathbf{Q} \\ \mathbf{Q}\mathbf{Q}^T &= \mathbf{I} \end{aligned} \quad (4)$$

where $\mathbf{Q}$ is an orthogonal matrix. If $\mathbf{D}_s$ has different diagonal elements, $\mathbf{Q}$ will be a permutation matrix and we may conclude that the steering vectors in $\mathbf{U}_s$ are correlated estimated despite a change of the steering vector in the order. Unfortunately, this is not usually the case.

Step 2, calculate the higher order cumulants of the received signal, e.g. the four order cumulants, which is defined as [8]

$$\begin{aligned} \text{Cum}(x_i, x_j, x_k, x_l) &= E(x_i x_j x_k x_l) - E(x_i x_j)E(x_k x_l) \\ &\quad - E(x_i x_k)E(x_j x_l) - E(x_i x_l)E(x_j x_k) \end{aligned} \quad (5)$$

By fixing the last two variables in the expression and varying the first two variable as x_i and x_j , one may generate a cumulant matrix $\mathbf{C}(x_k, x_l)$ (where x_k and x_l can be altered to obtain a different matrix). Since the input sources are zero mean and independent variables, the cumulant matrix of the received signal $\mathbf{x}$ is related to the cumulant matrix of the input signal $\mathbf{s}$ by [8]

$$\begin{aligned} \mathbf{C}(x_k, x_l) &= \mathbf{B}\mathbf{K}_{lk}\mathbf{C}_s\mathbf{B}^T \\ \mathbf{C}_s &= \text{diag}(\text{cum}(s_1, s_1, s_1, s_1), \dots, \text{cum}(s_N, s_N, s_N, s_N)) \\ \mathbf{K}_{lk} &= \text{diag}(B_{k1}B_{l1}, \dots, B_{kN}B_{lN}) \end{aligned} \quad (6)$$

where $\mathbf{C}_s$ is a diagonal matrix relating to the fourth order cumulants of the input signals, $\mathbf{K}$ is a diagonal matrix relating to the product of the l th and k th rows of matrix $\mathbf{B}$.

Substituting Eq. (4) into Eq. (6), one gets

$$\text{inv}\left(\sqrt{\mathbf{D}_s}\right) \mathbf{U}_s^T \mathbf{C}(x_k, x_l) \mathbf{U}_s \text{inv}\left(\sqrt{\mathbf{D}_s}\right) = \mathbf{Q} \mathbf{K}_{lk} \mathbf{C}_s \mathbf{Q}^T \quad (7)$$

Change k and l , one is able to obtain a set of matrix equations analogues to Eq. (7). Combining them linearly, we have the following matrix [8]

$$\sum_k \sum_l \alpha_{lk} \mathbf{C}(x_k, x_l) = \mathbf{Q} \sum_k \sum_l \alpha_{lk} \mathbf{K}_{lk} \mathbf{C}_s \mathbf{Q}^T \quad (8)$$

By carefully choosing the value of α_{lk} , one is able to obtain a matrix with distinct eigen values [8]. According to the matrix theory, such matrixes will have distinct eigen vectors which are orthogonal to each other. Therefore, if we perform eigen value decomposition of the matrix in Eq. (8), matrix $\mathbf{Q}$ will be unique (despite multiplication by a permutation matrix) and can be used for the evaluation of the matrix $\mathbf{B}$ in Eq. (4).

The key burden of application of such an algorithm to estimate the steering vectors in antenna arrays is the complex nature of the received signals. The complex nature will impact the uniqueness of matrix $\mathbf{Q}$. If the matrix is multiplied by a complex factor with a unit modulus, it still fulfills Eq. (8).

There are two ways to resolve this problem. Firstly, we may decompose the matrix in Eq. (8) and calculate the corresponding rows of $\mathbf{Q}$ to estimate the steering vector. However, the true steering vector should be the estimated one multiplied by a complex factor with a unit modulus. The phase of the complex number can be estimated afterwards together with the DOA and the mutual coupling matrix.

The other way is to convert the complex matrix into the real one by separating the real part and imaginary part of the matrix as well as the signals. In such a case, Eq. (1) can be reformulated as

$$\begin{pmatrix} \text{Re } \mathbf{x} \\ \text{Im } \mathbf{x} \end{pmatrix} = \begin{pmatrix} \text{Re } \mathbf{B} & -\text{Im } \mathbf{B} \\ \text{Im } \mathbf{B} & \text{Re } \mathbf{B} \end{pmatrix} \begin{pmatrix} \text{Re } \mathbf{s} \\ \text{Im } \mathbf{s} \end{pmatrix} + \begin{pmatrix} \text{Re } \mathbf{n} \\ \text{Im } \mathbf{n} \end{pmatrix} \quad (9)$$

where Re and Im denote the real part and the imaginary part of the matrix and the vector. Considering the real part and the imaginary part of the signals are independent variables, which is usually the case in practical communication systems, one may estimate the matrix composed by the real part and the imaginary part of the matrix $\mathbf{B}$. Therefore, the steering vector of the antenna array can be estimated correctly despite a factor of -1 or exchange of the real and imaginary part of the steering vector.

III. ESTIMATION OF THE DOA AND THE MUTUAL COUPLING MATRIX BASED ON THE ESTIMATED STEERING VECTORS

As discussed in the previous section, the steering vectors can be blindly estimated by the higher order cumulants of the received signals. The estimated steering vector either differs

from the real steering vector by complex factor with unit modulus while using the complex higher order cumulants, and by a factor of -1 or exchange of the real and imaginary part of the steering vector while using the real channel matrix model in Eq. (9). These unknown factors are denoted by a parameter set of φ .

While jointly estimating the DOA and the mutual coupling matrix $\mathbf{C}$, the parameter φ should be included in the estimation process. Denote the estimated steering vector matrix as $\hat{\mathbf{B}}$, the target is to obtain the minimized square error function while adjusting the parameters like φ , DOA $\boldsymbol{\theta}$, and the coupling matrix $\mathbf{C}$.

$$\begin{aligned} J &= \|\hat{\mathbf{B}}(\varphi) - \mathbf{C}\mathbf{A}\| \\ \{\varphi, \mathbf{C}, \boldsymbol{\theta}\} &= \arg \min_{\varphi, \mathbf{C}, \boldsymbol{\theta}} J \\ \boldsymbol{\theta} &= (\theta_1, \theta_2 \dots \theta_N) \end{aligned} \quad (10)$$

The estimation can be decomposed into three steps with each step estimating one set of the parameters. It was revealed in [8], since the coupling matrix can be considered as a Toeplitz matrix, $\mathbf{C}$ can be obtained analytically using the least square method (LSM).

During each step, the target function J will be minimized. Hence, the target function J will always decrease. The iteration will finally converge to an optimal solution which corresponds to the correct DOA and antenna mutual coupling matrix.

IV. SIMULATION RESULTS AND DISCUSSIONS

Without loss generality, we perform our algorithm on a four element linear antenna array. The antenna elements are dipole antennas with the length of $\lambda/2$ and the distance between the adjacent elements is $\lambda/2$. The mutual coupling admittance matrix is calculated using the method of moment (MOM) by assuming that the circuits are short circuited and the mutual coupling impedance matrix is calculated by inverting the admittance matrix. The mutual impedance matrix is (with the units of Ω):

$$\begin{pmatrix} 89.2877 + 45.7768i & -22.7798 - 32.7712i & 11.0926 + 20.9159i & -6.8773 - 15.0432i \\ -22.7799 - 32.7711i & 90.6461 + 46.2778i & -23.3408 - 33.0217i & 11.0926 + 20.9159i \\ 11.0927 + 20.9159i & -23.3408 - 33.0216i & 90.6461 + 46.2778i & -22.7798 - 32.7712i \\ -6.8774 - 15.0432i & 11.0927 + 20.9159i & -22.7799 - 32.7711i & 89.2877 + 45.7768i \end{pmatrix} \quad (11)$$

The coupling matrix is related to the impedance matrix by

$$\mathbf{C} = \left(\frac{Z_L \mathbf{I} + \mathbf{Z}}{Z_L} \right)^{-1} \quad (12)$$

where Z_L is the load resistance of the antenna element, $\mathbf{I}$ the identity matrix. In the simulation, Z_L is assumed to be 75Ω .

Assuming that the signals are coming in the direction of -20 degree and +20 degree with the unit power and SNR is 10dB, the corresponding steering matrix $\mathbf{B}$ should be

$$\begin{pmatrix} 0.3918 - 0.0530i & 0.4437 - 0.1560i \\ -0.1681 + 0.4484i & -0.2800 - 0.3472i \\ -0.5562 - 0.0133i & -0.5584 + 0.0762i \\ -0.2873 - 0.4258i & -0.1435 + 0.4502i \end{pmatrix} \quad (13)$$

Using the algorithm described in the above sections, the steering matrix can be estimated as

$$\begin{pmatrix} 0.3876 - 0.0519i & 0.4474 - 0.1583i \\ -0.1686 + 0.4511i & -0.2714 - 0.3437i \\ -0.5510 - 0.0139i & -0.5638 + 0.0795i \\ -0.2827 - 0.4295i & -0.1563 + 0.4499i \end{pmatrix} \quad (14)$$

and the estimated coupling matrix $\mathbf{C}$ is

$$\begin{pmatrix} 0.4322-0.1124i & 0.0857+0.0270i & -0.0409-0.0205i & 0.0207-0.0028i \\ 0.0857+0.0270i & 0.4322-0.1124i & 0.0857+0.0270i & -0.0409-0.0205i \\ -0.0409-0.0205i & 0.0857+0.0270i & 0.4322-0.1124i & 0.0857+0.0270i \\ 0.0207-0.0028i & -0.0409-0.0205i & 0.0857+0.0270i & 0.4322-0.1124i \end{pmatrix} \quad (15)$$

It is very close to the real coupling matrix $\mathbf{C}$ calculated by Eq. (12) with less than 5% deviation for each element of the matrix.

$$\begin{pmatrix} 0.4368-0.0916i & 0.0848+0.0376i & -0.0373-0.0105i & 0.0246 + 0.0071i \\ 0.0848+0.0376i & 0.4423-0.0752i & 0.0824+0.0325i & -0.0373 - 0.0105i \\ -0.0373-0.0105i & 0.0824+0.0325i & 0.4423-0.0752i & 0.0848 + 0.0376i \\ 0.0246+0.0071i & -0.0373-0.0105i & 0.0848+0.0376i & 0.4368 - 0.0916i \end{pmatrix} \quad (16)$$

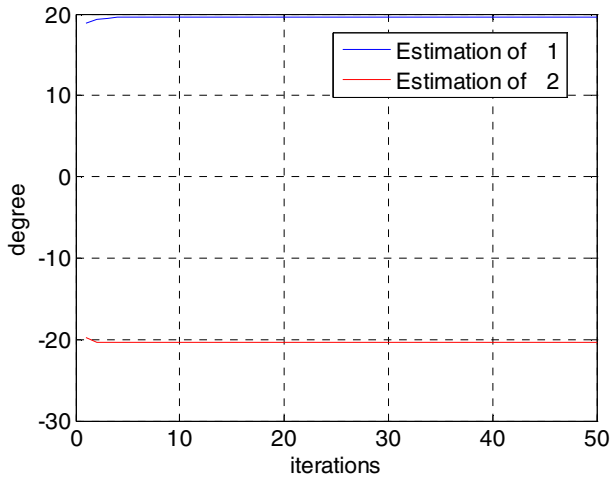


Fig. 1. Estimated DOAs of the two signal sources

For the DOAs, the estimated results versus iteration number are plotted in Fig. 1. It can be seen that the DOAs are correctly estimated with the value very close to 20 degree and -20 degree.

The small deviation comes from the fact that the coupling matrix Eq. (12) is not a perfect Toeplitz matrix.

To summarize, the DOAs and the mutual coupling matrix are correctly estimated using the algorithm proposed in the previous sections.

V. CONCLUSION

We have proposed a novel algorithm for the blind estimation of the signal DOA and antenna mutual coupling matrix. The blind estimation is realized by evaluating the higher order cumulants of the received signals. The parameters relating to the true steering vectors of the antenna array, such as the DOA and the mutual coupling matrix, can be obtained afterwards by minimizing the error between the steering matrix obtained by the cumulants based algorithm and the matrix estimated by the guessing values of the DOA and the mutual coupling matrix. Numerical simulations are performed to verify the proposed algorithm and very good performance is obtained during the estimation of the DOA and mutual coupling matrix for a four element antenna array.

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